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COURSE CODE: CSA1401

LAB:

1.The Company ABC runs with employees with several departments. The Organization manager had all the mobile numbers of employees. Assume that you are the manager and need to verify the valid mobile numbers because there may be some invalid numbers present. Implement a LEX program to check whether the mobile number is valid or not.

CODE:

```
%{
%}

%%

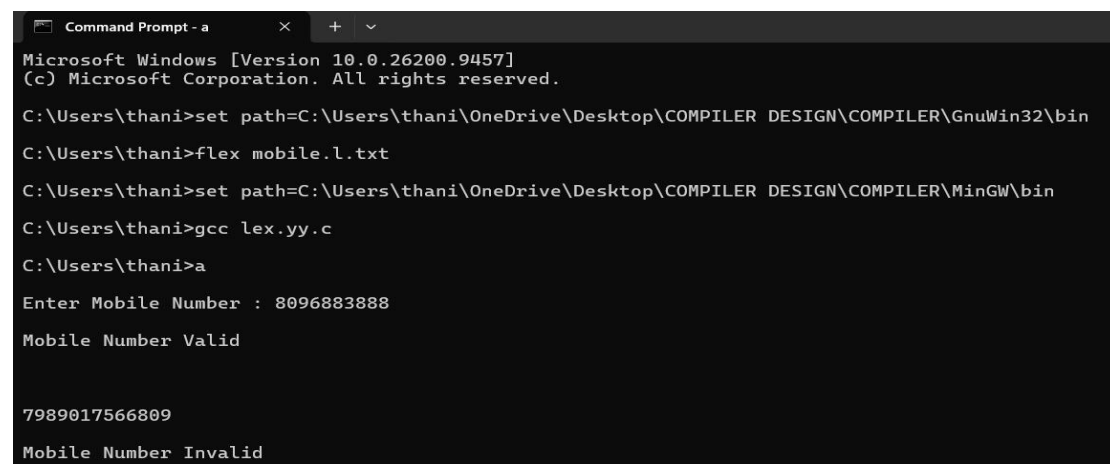
[6-9][0-9]{9} {printf("\nMobile Number Valid\n");}

.+ {printf("\nMobile Number Invalid\n");}

%%
int yywrap(void) {}

int main()
{
    printf("\nEnter Mobile Number : ");
    yylex();
    printf("\n");
}
```

OUTPUT:



```
Command Prompt - a
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\thani>set path=C:\Users\thani\OneDrive\Desktop\COMPILER DESIGN\COMPILER\GnuWin32\bin
C:\Users\thani>flex mobile.l.txt
C:\Users\thani>set path=C:\Users\thani\OneDrive\Desktop\COMPILER DESIGN\COMPILER\MinGW\bin
C:\Users\thani>gcc lex.yy.c
C:\Users\thani>a
Enter Mobile Number : 8096883888
Mobile Number Valid

7989017566809
Mobile Number Invalid
```

2. Write a LEX Program to check the email address is valid or not.

CODE:

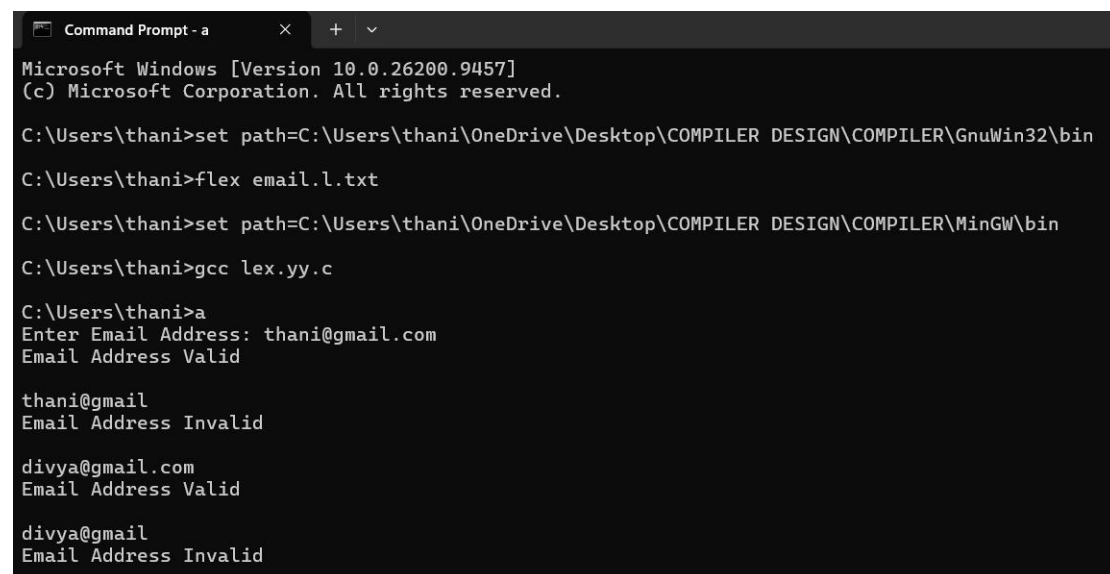
```
%{
#include <stdio.h>
%}

%%
^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$ { printf("Email Address
Valid\n"); }
.* { printf("Email Address Invalid\n"); }
%%

int yywrap()
{
    return 1;
}

int main()
{
    printf("Enter Email Address: ");
    yylex();
    return 0;
}
```

OUTPUT:



```
Command Prompt - a
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\thani>set path=C:\Users\thani\OneDrive\Desktop\COMPILER DESIGN\COMPILER\GnuWin32\bin
C:\Users\thani>flex email.l.txt
C:\Users\thani>set path=C:\Users\thani\OneDrive\Desktop\COMPILER DESIGN\COMPILER\MinGW\bin
C:\Users\thani>gcc lex.yy.c
C:\Users\thani>a
Enter Email Address: thani@gmail.com
Email Address Valid

thani@gmail
Email Address Invalid

divya@gmail.com
Email Address Valid

divya@gmail
Email Address Invalid
```

3. Write a LEX program to check whether the given input is a digit or not.

CODE:

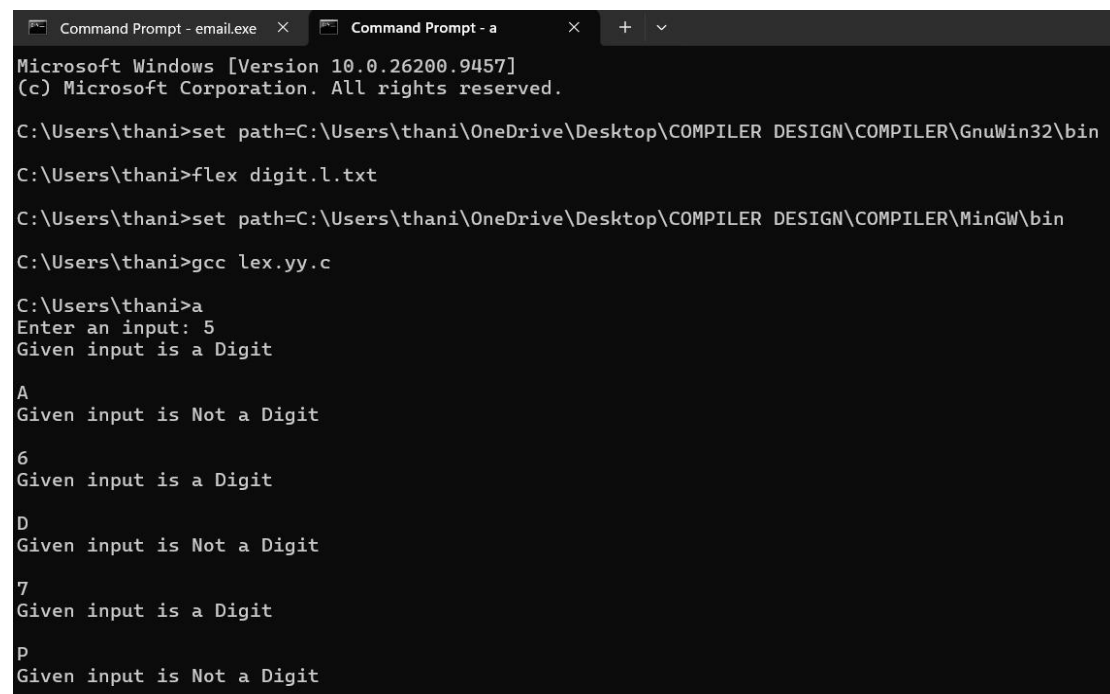
```
%{
#include <stdio.h>
%}

%%
^[0-9]$  { printf("Given input is a Digit\n"); }
.*      { printf("Given input is Not a Digit\n"); }
%%

int yywrap()
{
    return 1;
}

int main()
{
    printf("Enter an input: ");
    yylex();
    return 0;
}
```

OUTPUT:



```
Command Prompt - email.exe  Command Prompt - a  +  v
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\thani>set path=C:\Users\thani\OneDrive\Desktop\COMPILER DESIGN\COMPILER\GnuWin32\bin
C:\Users\thani>flex digit.l.txt
C:\Users\thani>set path=C:\Users\thani\OneDrive\Desktop\COMPILER DESIGN\COMPILER\MinGW\bin
C:\Users\thani>gcc lex.yy.c
C:\Users\thani>a
Enter an input: 5
Given input is a Digit

A
Given input is Not a Digit

6
Given input is a Digit

D
Given input is Not a Digit

7
Given input is a Digit

p
Given input is Not a Digit
```